

APPLICATIONS OF THE SVD

Eckart-Young theorem $\rightsquigarrow$ tells us how to construct an optimal low-rank approximation of $A \in \mathbb{R}^{m \times n}$

$$A = U \Sigma V^T \quad \text{singular value decomposition}$$

$$A_k = \begin{bmatrix} | & | & | \\ U_k & \Sigma_k & V_k^T \\ | & | & | \end{bmatrix} \begin{matrix} \uparrow \\ \text{first } k \text{ columns of } U \end{matrix} \begin{matrix} \uparrow \\ \text{top-left } k \times k \text{ submatrix of } \Sigma \end{matrix} \begin{matrix} \rightarrow \\ \text{first } k \text{ columns of } V \end{matrix}$$

$$\rightarrow A_k \in \arg\min_{\text{rank}(B) \leq k} \|A - B\|_{2,F}$$

$\downarrow$
both for Spectral and Frobenius norm

$$A = \begin{bmatrix} | & | & | \\ U & \Sigma & V^T \\ | & | & | \end{bmatrix} \quad \sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0$$

$$A = \sigma_1 \cdot u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_n u_n v_n^T \quad u_i v_i^T = \begin{bmatrix} | & | \\ \hline \end{bmatrix} = \begin{bmatrix} | & | \end{bmatrix}$$

Examples of why it's helpful to have a low-rank approximation:

① storage cost

$$n \times n \text{ matrix} \rightarrow n^2$$

$$A \approx \begin{bmatrix} | & | & | \\ \hline \end{bmatrix} \begin{bmatrix} | & | & | \\ \hline \end{bmatrix} \begin{bmatrix} | & | & | \\ \hline \end{bmatrix} \rightarrow \approx 2nk + k^2 \text{ numbers}$$

② fast operations:

$$n \begin{bmatrix} | & | & | \\ \hline \end{bmatrix} \begin{bmatrix} | & | & | \\ \hline \end{bmatrix} \begin{bmatrix} | & | & | \\ \hline \end{bmatrix} \rightarrow O(n^2)$$

$$n \begin{bmatrix} | & | & | \\ \hline \end{bmatrix} \begin{bmatrix} | & | & | \\ \hline \end{bmatrix} \begin{bmatrix} | & | & | \\ \hline \end{bmatrix} \begin{bmatrix} | & | & | \\ \hline \end{bmatrix} \rightarrow O(nk)$$

$\uparrow$
vector

Example: Image



$\rightsquigarrow$ matrix

Frobenius norm:

$$\|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \dots + \sigma_n^2} \quad \text{best rank-}k \text{ approx}$$

Spectral norm:

$$\|A - A_k\|_2 = \sigma_{k+1}$$

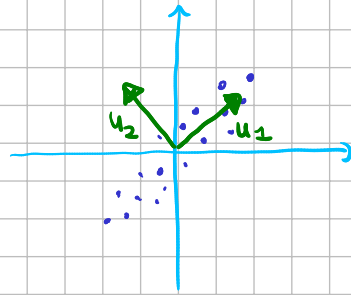
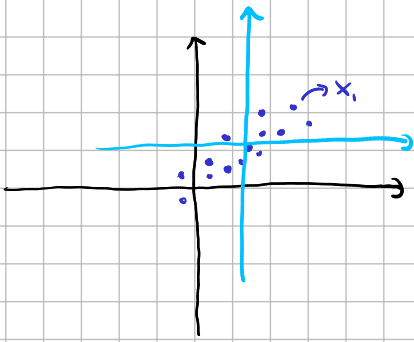
PRINCIPAL COMPONENT ANALYSIS

Assume we have data $x_1, x_2, \dots, x_n \in \mathbb{R}^m \rightsquigarrow A = \begin{bmatrix} | & | & | \\ x_1 & x_2 & \dots & x_n \\ | & | & | \end{bmatrix} \in \mathbb{R}^{m \times n}$

De-mean the data: compute $\mu := \frac{1}{n} (x_1 + x_2 + \dots + x_n) \in \mathbb{R}^m$

$$\rightsquigarrow \hat{x}_j = x_j - \mu \rightsquigarrow \hat{A} = \begin{bmatrix} | & | & | \\ \hat{x}_1 & \hat{x}_2 & \dots & \hat{x}_n \\ | & | & | \end{bmatrix} \in \mathbb{R}^{m \times n}$$

if $\hat{A} = U \Sigma V^T \rightarrow$ the vectors $u_1, u_2, \dots$ are called the "principal components" of our data.



u_1 = first principal component
 = left sing. vector of $\hat{A}$ corresponding to the largest singular value
 = direction in which our data has the maximum variance.

$$\text{if } \hat{A} \approx u_1 \cdot \sigma_1 \cdot v_1^T = \begin{bmatrix} w_1 u_1 & w_2 u_1 & \dots & w_n u_1 \end{bmatrix}$$

$\downarrow$
 $v_1^T = [w_1, w_2, \dots, w_n]$

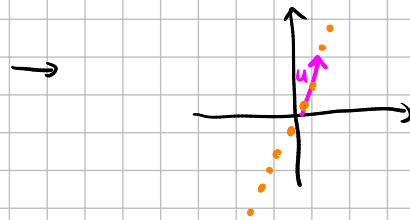
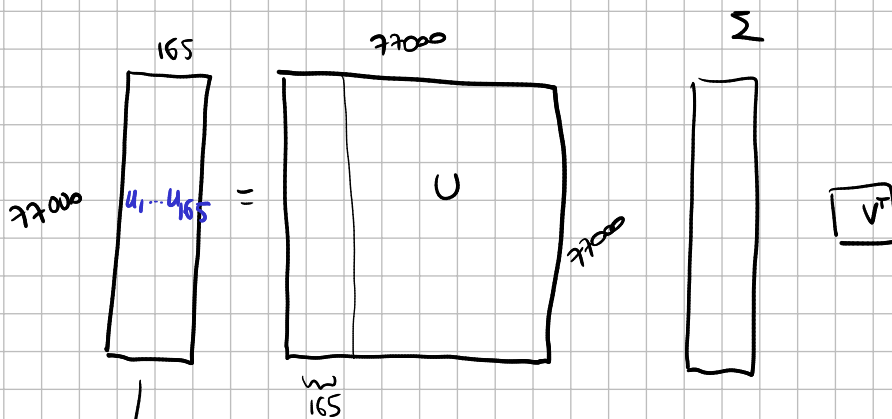
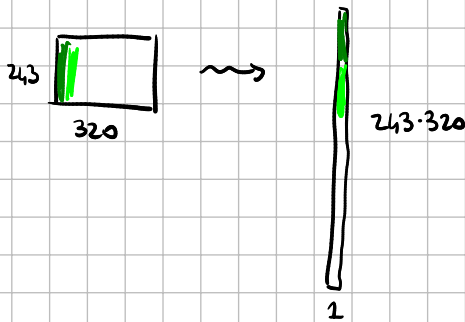


Image classification: Yale face dataset

15 people, 11 photos each (with different expressions)



$$= \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_{165} u_{165} v_{165}^T$$

$\leadsto$ we don't care about $u_{166}, \dots, u_{7700}$

$u_1, u_2, \dots, u_{165}$ principal components

J-th column
↓

$x = \text{one face}$

$$\leadsto x = \mu + \alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_{165} u_{165}$$



$$\hat{A} = \sigma_1 u_1 v_1^T + \dots + \sigma_{165} u_{165} v_{165}^T$$

$$\Rightarrow \hat{A}(:, j) = u_1 \cdot (\sigma_1 v_{1j}) + u_2 \cdot (\sigma_2 v_{2j}) + \dots + u_{165} \cdot (\sigma_{165} v_{165j})$$

Data visualization: data points in $\mathbb{R}^{77000} \leadsto$ project them onto $\mathbb{R}^2$

Image classification:

each image j in the training set $\leadsto$ has coeff. $\alpha_1^j, \alpha_2^j, \dots, \alpha_{165}^j$

Given a new image, I can compute its coefficients

$$\beta_1, \beta_2, \dots, \beta_{165}$$

Actually, only consider a subset of principal components

- 1) compute all the coefficients ("scores") of the training images wrt these components
- 2) compute the coeff ("scores") of the test image wrt to the components
- 3) Associate, to the test image, the training image with the closest scores.